

Algebraic Intermediate Representation (AIR) = three ingredients

Public inputs

values known to both
prover and verifier
(e.g. start x, result y)

Transition constraints

polynomials that are 0 on
every legal step and
non-zero on any illegal one

Boundary assertions

tie public inputs to
specific cells of the trace
(first / last rows)

A pair (current state, next state) is valid iff every transition constraint vanishes on it.